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The Pinski–Narin influence weight and the Ramanujacharyulu power-weakness ratio indicators revisited

Gangan Prathap[1]

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Abstract

A graph theoretic approach from social network analysis allows size-dependent and size-independent bibliometric indicators to be identified from what is called the citation matrix. In an input–output sense, the number of references becomes the size-dependent measure of the input and the number of citations received by the journal from all journals in the network becomes the size-dependent measure of the output. However, in this paper, we are interested to compare two size-independent dimensionless indicators: the Pinski–Narin influence weight (IW) and the Ramanujacharyulu power-weakness ratio (PWR). These are proxies for the quality of the journal's performance in the network. We show that at the non-recursive level, the two indicators are identical. At this stage these are simply measures of popularity. After recursion (i.e. repeated improvement or iteration) these become network measures of prestige of the journals. PWR is computed as a ratio of terms in the weighted citations vector and weighted references vector after the power and weakness matrices are separately recursively iterated. The Pinski–Narin procedure computes a matrix of ratios first and evaluates the IW after recursive iteration of this matrix of ratios. In this sense, the two procedures differ just like the RoA (ratio of averages) and the AoR (average of ratios) ways of computing relative citation indicators. We illustrate the concepts using datasets from subgraphs of 10 statistical journals and 14 Chinese chemistry journals with network data collected from the Web of Science. We are also able to show the confounding effects when self-citations are taken into account.

Keywords Bibliometrics · Social network analysis · Journal performance metrics · Indicators · Quality · Quantity · Power-weakness ratio · Influence weight

✉ Gangan Prathap
gangan_prathap@hotmail.com

[1] A P J Abdul Kalam Technological University, Thiruvananthapuram, Kerala 695016, India

Introduction

Prathap (2018) showed that citation analysis as a tool, first for journal evaluation, and later for more broad-based research evaluation of individuals, groups, institutions and scientists can be interpreted in terms of two Aristotelian categories—quality and quantity (Garfield and Sher 1963; Garfield 1972). The size of the journal belonging to a network (say that in the Web of Science Core Collection), is measured by the count of all articles P published in the journal during a chosen window (called the publications window). This is a size-dependent input term for evaluating journal performance. The size-dependent output term is the number of citations C received by these P articles from all articles published in all journals in the network during a specified period called the citations window. From these it is possible to derive a size-independent quality proxy called impact $i = C/P$. This was the original Journal Impact Factor, the numerator C being the number of cites in the current year (citations window) to the articles published in the previous 2 years (publications window) while the numerator is the number of articles P published during the same publications window. Note that this factor is now a size-independent ratio of two size-dependent values but is actually not dimensionless; it has the dimensions of P (Prathap 2018).

Several new indicators for journal performance evaluation (Bollen et al. 2006; Leydesdorff 2009; West and Vilhena 2014) have now appeared based on what is called a network approach. One can see an early intimation of this in the use of journal to journal cross citations to measure the influence of one journal on another and one subfield on another by Cason and Lubotsky (1936). An interesting insight is that one can use this data to measure quantitatively the extent to which each psychological field influences and is influenced by each of the other psychological fields. In a way as we shall see later, this anticipates the extension of Ramanujacharyulu's "tournaments" metaphor (1964) to bibliometrics. This approach using tournament tables appears first in Kendall (1955) and is well discussed by David (1971).

Daniel and Louttit (1953) used a cross citing matrix in psychology to measure the similarity of the individual journal citation patterns and succeeded in obtaining a formal clustering of scientific journals. Similarly, Kessler (1964) formulated a journal cross citing matrix for physics journals and hypothesized that specific types of information could be deduced from this matrix. Xhignesse and Osgood (1967) extended network theory concepts to portray the relationships between journals and to measure their referencing similarities.

Pinski and Narin (1976) proposed an iterative algorithm based on a network approach. The citation matrix was first normalized and then an eigenvalue operation was used so that instead of a raw (non-recursive) count of citations C, a weighted (recursive) count will take cognizance of the "prestige" of the citing journal. In a subsequent section, we shall see how this is operationalized in further detail. These are problems that arise frequently in social network analysis and there are well-established graph theoretic tools that allow the computation of the recursive indicators. Prathap et al. (2016) exploited an idea that was introduced by Ramanujacharyulu (1964) for defining a new size-independent network property. The raw count of citations is a non-recursive non-network measure. Taking into account the "prestige" of the journal from which a citation arises is a network measure that needs a recursive, iterative computation (Pinski and Narin 1976; Brin and Page 2001; Bergstrom 2007). That is, while counting total citations is a non-recursive measure, weighting the count taking the "prestige" of the journal from which the citation arises into account needs a recursive, iterative computation. In a sense, the "raw" count of citations is a measure of "popularity" and "weighted" recursively computed counts of citations measure "prestige" (Bollen et al. 2006; Yan and Ding 2010).

Journal indicators can be size-dependent or size-independent metrics (Pinski and Narin 1976; De Visscher 2010, 2011, 2012). The Journal Impact Factor (Garfield and Sher 1963) is computed from two size-dependent indicators: the total number of articles P published in the journal during a 2-year publication window and the total citations C to these articles from all articles published during the 1-year citation window immediately following the publication window. For example, impact factor for any journal for the year 2013 has 2011 and 2012 as the publication window. Then the impact i is computed as citations per paper C/P, which is size-independent measure but not dimensionless. That is, while C is a size-dependent total impact, i is a size-independent specific impact that has the dimensions of P. The impact factor, i began to be used as a proxy for quality; P is the candidate proxy for quantity or size.

In all this, dimensionality plays a crucial role. Bibliometric indicators that come from social network considerations may measure different dimensions of the citation networks, or may be highly correlated among themselves (Leydesdorff 2009). The two main dimensions—size and impact, together shape a property called "influence" (Bensman 2007; Leydesdorff 2009). The primary non-recursive non-network indicators like P, C and i, have dimensions like size/quantity assigned to P and quality assigned to i, while C being a product of quantity and size has both dimensions and can be identified with total impact or total "influence" of a journal. It is not so easy to assign dimensionality to the recursive network indicators that emerge from a graph-theoretical and social network methodology. This is clear from the evaluation of bibliometric indicators arising from the social network analysis performed on journal citation data by Leydesdorff (2009) and Bollen et al. (2009).

In the following sections of this paper we focus attention on two dimensionless size-independent metrics for journal evaluation that arise naturally from a network approach. One is the Pinski and Narin (1976) influence weight (IW). The other is the Ramanujacharyulu power-weakness ratio (1964). To illustrate the behavior of these indicators to perform as dimensionless size-independent journal metrics, we choose two real life journal ecosystems, a subgraph of leading statistical journals and another of leading Chinese chemistry journals.

The Ramanujacharyulu power-weakness ratio (PWR) and the Pinki-Narin influence weight (IW)

Ramanujacharyulu (1964), working with the associated matrices that arise in graph theory, proposed to balance the "power to influence" with the "weakness to be influenced" through a measure called the power-weakness ratio. That this "tournaments" metaphor (Leydesdorff et al. 2016) could be used in bibliometric studies was anticipated much earlier by Cason and Lubotsky (1936). Table 1 is an example of a "cited-citing" matrix that arises in a bibliometric formulation. We shall follow the terminology used to compute the Eigenfactor Score and the Article Influence Score indicators to explain the principal features (Bergstrom 2007; West et al. 2010). Let **Z** be the cited-citing matrix. If the entries are read row-wise, then for a journal in row i, an entry such as z_{ij} are the citations from journal j in the citation window (say 2013) to articles published in journal i during the publications window (say 2011–2012); in social network analysis these are the in-coming links. It is important to emphasize here that the matrix can also be read column-wise; now for the

Table 1 The **Z** matrix of the cross citing terms of the ten statistical journals as a subgraph of the main graph of all the journals in the Web of Science Core Collection

Statistics Journals in JCR (Power matrix)	Am Stat	Ann Stat	Economet Theor	Econometrica	J Am Stat Assoc	J Comput Graph Stat	J Roy Stat Soc A Sta	J Roy Stat Soc B	Scand J Stat	Technometrics	Citations
Am Stat	**42**	3	1	2	11	23	5		1	4	92
Ann Stat	8	**621**	105	34	302	94	7	113	144	21	1449
Economet Theor	1	25	**140**	20	15	1	4	12	24		242
Econometrica	4	58	135	**431**	92	5	19	25	18	1	788
J Am Stat Assoc	48	228	59	18	**460**	139	35	108	70	63	1228
J Comput Graph Stat	8	20			33	**66**		8	10	18	163
J Roy Stat Soc A Sta	5	7	1	1	14	3	**57**	5	6	2	101
J Roy Stat Soc B	12	118	18	4	195	68	9	**81**	56	39	600
Scand J Stat	2	24	5	1	31	13		11	**48**	3	138
Technometrics	15	11	1		27	21	2	5	6	**108**	196
References	145	1115	465	511	1180	433	138	368	383	259	4997

journal in column j, the entry z_{ij} are the references from journal j in the citation window (2013) to articles published in journal i during the publications window (2011–2012). In social network analysis these are the out-going links. Thus, row-wise, we see the journal i's "power to influence" and "column-wise" we see the journal j's "weakness to be influenced." The row-sum corresponding to row i is therefore the non-recursive indicator C, i.e. the total citations to journal i from all the journals in the ecosystem, including itself. This is taken as a measure of the "popularity" of journal i. If we also have an article vector $\boldsymbol{a}$, where a_i is the number of articles published by journal i over the publication window, then this is the value P for journal i and the ratio C/P is the non-recursive impact of the journal. A note of caution to be introduced here is that the subscript i is used here as an indicial notation and elsewhere in this paper, from the compulsions of historical legacy, as the notation for journal impact.

In the graph theoretic sense, $\mathbf{Z}=[\mathbf{Z_{ij}}]$ is the matrix associated with the graph (Ramanujacharyulu 1964). Many properties of such matrices are known and it can be raised indefinitely to the kth power, i.e. $\mathbf{Z}^{\mathrm{k}}$. This is the matrix used to define the "power of the journal to influence" (Ramanujacharyulu 1964). The Eigenfactor approach is thus a recursive iteration that raises $\mathbf{Z}$ to an order were convergence is obtained for what is effectively the weighted value of the total citations. So far the matrix calculations have all proceeded row-wise. For each journal we can find a value $p_i(k)$, which can be called the iterated power of order k of the journal i "to influence."

It is possible to carry out the same operations column-wise by using the transpose of the matrix $\mathbf{Z}^{\mathrm{T}}$ and then proceeding row-wise on these transposed elements in the same recursive and iterative manner indicated above (Ramanujacharyulu 1964). This now defines the "weakness of the journal to be influenced by." Again, for each journal we can find a value $w_i(k)$, which can be called the iterated weakness of order k of the journal i "to be influenced by."

At this stage we have two vectors of power k—the power vector $p(k)$ and the weakness vector $w(k)$. The elements of the former are the recursive counts of citations. The Eigenfactor methodology divides $p_i(k)$ by a_i, the number of articles published by journal i over the publication window, to get the Article Influence, which is the surrogate for the recursive impact of the journal. Prathap et al. (2016) proposed that instead of a_i, we take $w_i(k)$ as the recursive surrogate of the size of the journal. Then Ramanujacharyulu's power-weakness ratio of order k, $r_i(k)=p_i(k)/w_i(k)$ becomes a candidate for a size-independent recursive network measure of impact or quality of the journal. As $k\rightarrow\infty$, we get the converged power-weakness ratio. An elaboration of this procedure in formal terms was provided by Todeschini et al. (2015, p. 330).

At this stage it is meaningful to introduce the Pinski–Narin variant. Unlike Ramanujacharyulu (1964), Pinski and Narin (1976) does not deal with the eigenvectors from two matrices, namely the power and weakness forms, where the latter is the transpose of the former. Instead it normalises each row (the citation or cited-by data) of the power matrix by the total of the references of the journal corresponding to that row (i.e. the column sum). That is, Pinski and Narin (1976) normalized first the citations of a journal by dividing by the aggregated total number of ("citing") references along the column vector. Procedurally, for the entries in row i of the citation matrix (see Table 1), each value for that journal is normalized by the sum total of references of the column i corresponding to that journal. If there are N journals in all, in formulaic terms, one can write $\hat{z}_{ij}=z_{ij}/z_{+i}$ where $z_{+i}=\sum z_{ki}$ is summed column-wise over $k=1$ to N.

In this way, Pinksi and Narin will come up with a cross matrix of power-weakness ratios. Table 2 shows the normalized $\mathbf{Z}$ matrix of the cross power-weakness ratios of the

Table 2 The normalized **Z** matrix of the cross power-weakness ratios of the ten statistical journals as a subgraph of the main graph of all the journals in the Web of Science Core Collection

Statistics Journals in JCR (Power matrix)	Am Stat	Ann Stat	Economet Theor	Econometrica	J Am Stat Assoc	J Comput Graph Stat	J Roy Stat Soc A Sta	J Roy Stat Soc B	Scand J Stat	Technometrics	Citations
Am Stat	**0.29**	0.02	0.01	0.01	0.08	0.16	0.03	0	0.01	0.03	0.63
Ann Stat	0.01	**0.56**	0.09	0.03	0.27	0.08	0.01	0.1	0.13	0.02	1.30
Economet Theor	0	0.05	**0.3**	0.04	0.03	0	0.01	0.03	0.05	0	0.52
Econometrica	0.01	0.11	0.26	**0.84**	0.18	0.01	0.04	0.05	0.04	0	1.54
J Am Stat Assoc	0.04	0.19	0.05	0.02	**0.39**	0.12	0.03	0.09	0.06	0.05	1.04
J Comput Graph Stat	0.02	0.05	0	0	0.08	**0.15**	0	0.02	0.02	0.04	0.38
J Roy Stat Soc A Sta	0.04	0.05	0.01	0.01	0.1	0.02	**0.41**	0.04	0.04	0.01	0.73
J Roy Stat Soc B	0.03	0.32	0.05	0.01	0.53	0.18	0.02	**0.22**	0.15	0.11	1.63
Scand J Stat	0.01	0.06	0.01	0	0.08	0.03	0	0.03	**0.13**	0.01	0.36
Technometrics	0.06	0.04	0	0	0.1	0.08	0.01	0.02	0.02	**0.42**	0.76
References	0.50	1.46	0.79	0.97	1.84	0.85	0.56	0.59	0.65	0.69	8.89

ten statistical journals as a subgraph of the main graph of all the journals in the Web of Science Core Collection. At the zeroth-iteration, the sum of each row is the influence weight and a little reflection will show that this is the value of the Ramanujacharyulu PWR at the non-recursive (i.e. non-network) stage. Subsequently, after the recursive iteration, the IW will converge to values close to but not the same as the PWR after recursion. We are thus in the same dilemma as the famous AoR (Average of Ratios) vs RoA (Ratio of Averages) debate on relative citation indicators (see, e.g., Glänzel and Schubert 2018; Leydesdorff and Opthof 2018; Larivère and Gingras 2011). Do we trust the PWR or the IW? That is, do we perform the recursive iteration first on the power and weakness matrices independently and then take a ratio of the two eigenvectors, or do we take a matrix of power-weakness ratios and then find the eigenvector of the ratios after recursive iteration? We shall see later that there are interesting differences that are accentuated when self-citations are taken into consideration.

We illustrate the concepts using datasets from a subgraph of 10 statistical journals and 14 Chinese chemistry journals with network data collected from the Web of Science which was used to demonstrate the properties of the PWR indicator (Prathap et al. 2016; Prathap and Nishy 2016).

Real life applications of the power-weakness ratio (PWR) and the Pinski–Narin influence weight (IW) indicators

We now turn to real life examples to illustrate the behaviour of the power-weakness ratio (PWR) and Pinski–Narin influence weight (IW) as recursive size-independent dimensionless measures. The first journal ecosystem we choose comprises ten statistical journals from the Web of Science Core Collection (Prathap et al. 2016). The citation window is the year 2013 but the publication window will be all years preceding that. The matrix **Z** is set up easily with self-citations included, as is done in the Web of Science. Table 1 shows the **Z** matrix of the ten statistical journals as a subgraph of the main graph of all the journals in the Web of Science Core Collection. The weakness matrix is obtained as the transpose. These matrices are simple, irreducible and well-connected and the Ramanujacharyulu power and weakness iterations can be carried out using standard excel spreadsheets.

At the very first stage of the iteration, when $k=1$, we get the raw or non-recursive or non-network value of impact and when the iteration is continued to higher orders of k as $k \to \infty$ we found rapid convergence of the recursive or network power-weakness ratio and power weakness difference.

Table 3 displays the IW and PWR indicators of the ten statistical journals with self-citations before and after recursion (converged repeated iteration). This is graphically depicted in Fig. 1. Before recursion, the IW and PWR values are identical as predicted. From Table 3 and Fig. 1 we see that the popularity (before recursion) and prestige (after recursion and convergence) scores in each case are significantly different. Indeed, for Econometrica, this has nearly doubled. Also, IW and PWR converge to slightly different values.

It is easy to repeat this exercise if self-citations are to be excluded. This is done by inserting zero values in the diagonals of the citation matrix. Table 4 displays the IW and PWR indicators of the ten statistical journals without self-citations before and after recursion (converged repeated iteration). This is graphically depicted in Fig. 2. From Table 4 and Fig. 2 we see that the popularity (before recursion) and prestige (after

Table 3 The IW and PWR indicators of the ten statistical journals with self-citations as a subgraph of the main graph of all the journals in the Web of Science Core Collection before and after recursion (converged repeated iteration)

Statistics Journals in JCR	Non-recursive		Recursive	
	IW	PWR	IW	PWR
Am Stat	0.0713	0.0713	0.0328	0.0302
Ann Stat	0.1461	0.1461	0.1510	0.1461
Economet Theor	0.0585	0.0585	0.0470	0.0370
Econometrica	0.1734	0.1734	0.3571	0.3436
J Am Stat Assoc	0.1170	0.1170	0.1016	0.1064
J Comput Graph Stat	0.0423	0.0423	0.0247	0.0270
J Roy Stat Soc A Sta	0.0823	0.0823	0.0522	0.0607
J Roy Stat Soc B	0.1833	0.1833	0.1606	0.1690
Scand J Stat	0.0405	0.0405	0.0290	0.0308
Technometrics	0.0851	0.0851	0.0439	0.0493
Total	1.0000	1.0000	1.0000	1.0000

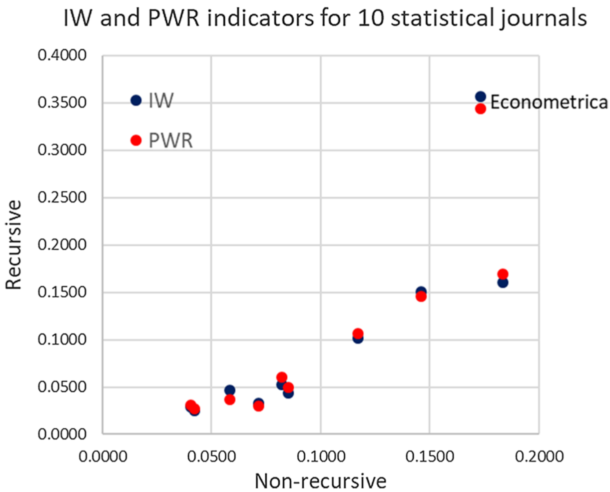


Fig. 1 IW and PWR indicators for 10 statistical journals with self-citations before and after recursion

recursion and convergence) scores are not so dramatically different. For Econometrica, which continues to be the dominant journal in this subgraph, the change is very small.

The second journal ecosystem we choose comprises fourteen Chinese chemistry journals from the JCR Web of Science Core Collection (Prathap and Nishy 2016). This is a highly localized ecosystem (i.e. subgraphs isolated from the global graph) and when the local ecosystem is isolated from the larger global scientific network, the cross-citation behavior within the local ecosystem reveals different features (Prathap and Nishy 2016). The power matrix **Z** for Chinese chemistry journals in JCR for citation window 2013 and publications window all years is given in Table 3 of Prathap and Nishy (2016) and not repeated here.

Table 5 displays the IW and PWR indicators of the fourteen Chinese chemistry journals with self-citations before and after recursion (converged repeated iteration). This is graphically depicted in Fig. 3. From Table 5 and Fig. 3 we see that the popularity (before recursion) and prestige (after recursion and convergence) scores are significantly

Table 4 The IW and PWR indicators of the ten statistical journals without self-citations as a subgraph of the main graph of all the journals in the Web of Science Core Collection before and after recursion (converged repeated iteration)

Statistics Journals in JCR	Non-recursive		Recursive	
	IW	PWR	IW	PWR
Am Stat	0.0423	0.0423	0.0323	0.0282
Ann Stat	0.1461	0.1461	0.1478	0.1422
Economet Theor	0.0274	0.0274	0.0475	0.0394
Econometrica	0.3890	0.3890	0.3719	0.3673
J Am Stat Assoc	0.0930	0.0930	0.0991	0.1015
J Comput Graph Stat	0.0230	0.0230	0.0239	0.0251
J Roy Stat Soc A Sta	0.0474	0.0474	0.0507	0.0574
J Roy Stat Soc B	0.1576	0.1576	0.1562	0.1650
Scand J Stat	0.0234	0.0234	0.0282	0.0305
Technometrics	0.0508	0.0508	0.0423	0.0434
Total	1.0000	1.0000	1.0000	1.0000

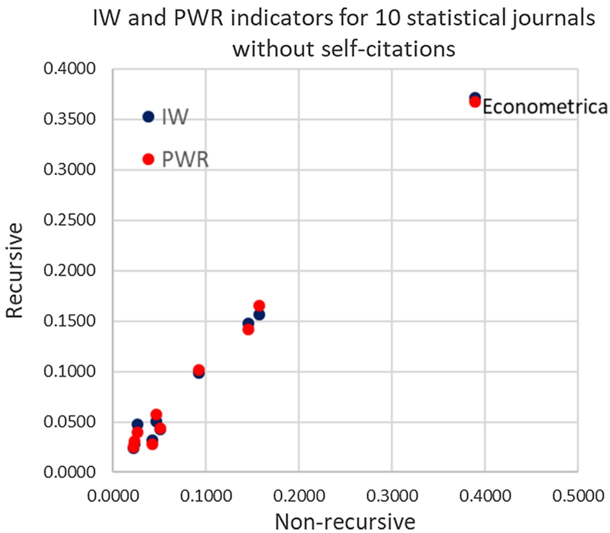


Fig. 2 IW and PWR indicators for 10 statistical journals without self-citations before and after recursion

different, before and after converged recursion in each case, and also from each other after recursion.

Once more, we repeat this exercise if self-citations are excluded. This is done by inserting zero values in the diagonals of the citation matrix. Table 6 displays the IW and PWR indicators of the fourteen Chinese chemistry journals without self-citations before and after recursion (converged repeated iteration). This is graphically depicted in Fig. 4. From Table 6 and Fig. 4 we see that the popularity (before recursion) and prestige (after recursion and convergence) scores are not so dramatically different, just as it was in Fig. 2 for the statistical journals. A noticeable difference between the two subgraphs is that for the statistical journals, there is considerable agreement between the IW and PWR indicators after recursion but this is not the case with the Chinese chemistry

Table 5 The IW and PWR indicators of the fourteen Chinese chemistry journals with self-citations as a subgraph of the main graph of all the journals in the Web of Science Core Collection before and after recursion (converged repeated iteration)

Chinese chemistry Journals in JCR	Non-recursive		Recursive	
	IW	PWR	IW	PWR
Acta chim sinica	0.1082	0.1082	0.1355	0.1714
Acta Phys-Chim Sin	0.0706	0.0706	0.0653	0.0719
Chem J Chinese U	0.0660	0.0660	0.0503	0.0511
Chem Res Chinese U	0.0658	0.0658	0.0469	0.0554
Chinese Chem Lett	0.0791	0.0791	0.1032	0.0601
Chinese J Anal Chem	0.0730	0.0730	0.0790	0.0598
Chinese J Catal	0.0741	0.0741	0.0791	0.0766
Chinese J Chem	0.0707	0.0707	0.0749	0.0763
Chinese J Inorg Chem	0.0537	0.0537	0.0229	0.0244
Chinese J Org Chem	0.0645	0.0645	0.0693	0.0493
Chinese J Struc Chem	0.0619	0.0619	0.0246	0.0268
J Rare Earth	0.0742	0.0742	0.1027	0.1490
Prog Chem	0.0663	0.0663	0.0640	0.0600
Sci China Chem	0.0719	0.0719	0.0820	0.0681
Total	1.0000	1.0000	1.0000	1.0000

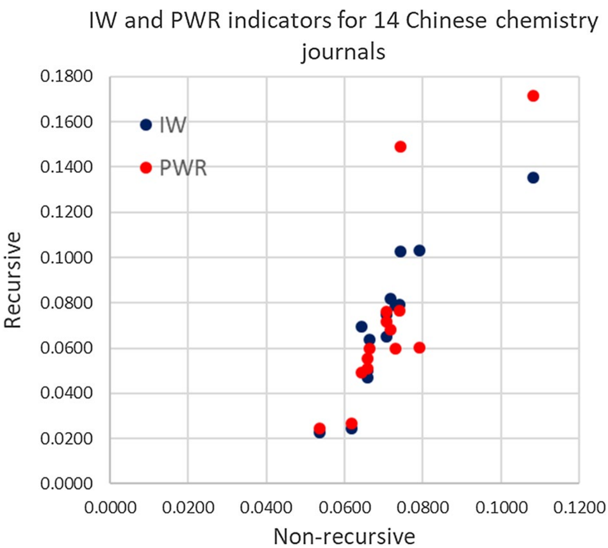


Fig. 3 IW and PWR indicators for 14 Chinese chemistry journals with self-citations before and after recursion

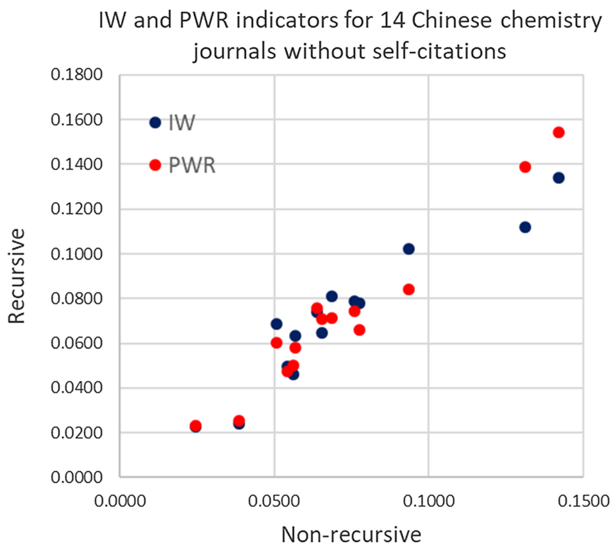
journals. The disagreement is greater when self-citations are included suggesting that self-citations can distort the measurement of journal to journal cross-citing behaviour.

A difficult question to answer is whether the IW or PWR is a better depiction of reality. In the former, the ratios are computed first within the matrix coefficients and then the eigenvalue recursion performed. In the latter case, the eigenvalue recursion is performed

Table 6 The IW and PWR indicators of the fourteen Chinese chemistry journals without self-citations as a subgraph of the main graph of all the journals in the Web of Science Core Collection before and after recursion (converged repeated iteration)

Chinese chemistry Journals in JCR	Non-recursive		Recursive	
	IW	PWR	IW	PWR
Acta Chim Sinica	0.1421	0.1421	0.1339	0.1542
Acta Phys-Chim Sin	0.0655	0.0655	0.0646	0.0710
Chem J Chinese U	0.0544	0.0544	0.0497	0.0475
Chem Res Chinese U	0.0564	0.0564	0.0463	0.0500
Chinese Chem Lett	0.0934	0.0934	0.1023	0.0840
Chinese J Anal Chem	0.0775	0.0775	0.0780	0.0661
Chinese J Catal	0.0759	0.0759	0.0790	0.0742
Chinese J Chem	0.0639	0.0639	0.0741	0.0758
Chinese J Inorg Chem	0.0246	0.0246	0.0227	0.0233
Chinese J Org Chem	0.0508	0.0508	0.0685	0.0603
Chinese J Struc Chem	0.0388	0.0388	0.0242	0.0254
J Rare Earth	0.1311	0.1311	0.1122	0.1388
Prog Chem	0.0568	0.0568	0.0635	0.0581
Sci China Chem	0.0687	0.0687	0.0810	0.0712
Total	1.0000	1.0000	1.0000	1.0000

Fig. 4 IW and PWR indicators for 14 Chinese chemistry journals without self-citations before and after recursion



separately on the power and weakness coefficients and the ratios are computed from the eigenvectors that result. The dilemma is exactly the same as that for another bibliometric conundrum, the debate that has become known as the "averages-of-ratios versus ratios-of-averages" debate (see, e.g., Glänzel and Schubert 2018; Leydesdorff and Opthof 2018; Larivère and Gingras 2011).

Concluding remarks

From graph theoretic considerations, we compare two size-independent dimensionless indicators: The Pinski–Narin influence weight (IW) and the Ramanujacharyulu power-weakness ratio (PWR). These are proxies for the quality of the journal's performance in the network. At the non-recursive level, the two indicators, which are non-network measures of popularity, are identical. After recursion (i.e. repeated improvement or iteration) these become network measures of prestige of the journals. PWR is computed as a ratio of weighted citations vector and weighted references vector after the power and weakness matrices are separately recursively iterated. The Pinski–Narin procedure computes a matrix of ratios first and evaluates the IW after recursive iteration of this matrix of ratios. In this sense, the two procedures differ just like the RoA (ratio of averages) and the AoR (Average of Ratios) ways of computing relative citation indicators.

We illustrate the concepts using datasets from subgraphs of 10 statistical journals and 14 Chinese chemistry journals with network data collected from the Web of Science. We are also able to show the distorting effects when self-citations are taken into account.

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